Solids, liquids and gases

1. Describe the position and motion of particles in each state of matter**.**

solid

.....

.....

liquid

.....

.....

gas

.....

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2. As you test substances, record your observations in the following table.

TEST	SOLID	LIQUID	GAS
shake container			
remove lid and pour onto table			
poke it			
change shape of container			
squash it			

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3. Which test could you use to show the difference between each of the following pairs of substances? Explain why it would work.

   a solid and a liquid

   .....

   .....

   a liquid and a gas

   .....

   .....

   a gas and a solid

   .....

   .....

4. Summarise properties of the three states of matter by completing the following table with ticks and crosses.

TEST	SOLID	LIQUID	GAS
fixed shape			
fixed volume			
compressible			

5. How do you think granulated sugar would behave in each of the following tests:

   Shake container.

   .....

   Remove lid and pour it.

   .....

   Poke it.

   .....

   Change shape of container.

   .....

   Squash it.

   .....

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6. What state of matter do you think granulated sugar is? Explain your choice.

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7. The learning object shows how a substance can change state when it is heated or cooled. What happens to the particles when they are:

heated

.....

.....

.....

cooled

.....

.....

.....

8. In the learning object you saw what happens to particles when they are heated. Use your knowledge to explain how two substances, with different boiling points, can be separated using evaporation. Use diagrams to help your explanation.

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9. A model has been used to show what particles in solids, liquids and gases look like. How is this model unrealistic? (There is more than one answer.)

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10. The learning object contains a real magnified photo of a solid on the last screen. How does its appearance compare with the model diagrams of a solid used in the animations?

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11. Why haven't scientists been able to take a photo of a gas?

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